

GRADE 9 MATHEMATICS · NUMBER

Powers & Exponents

THE BIG IDEA

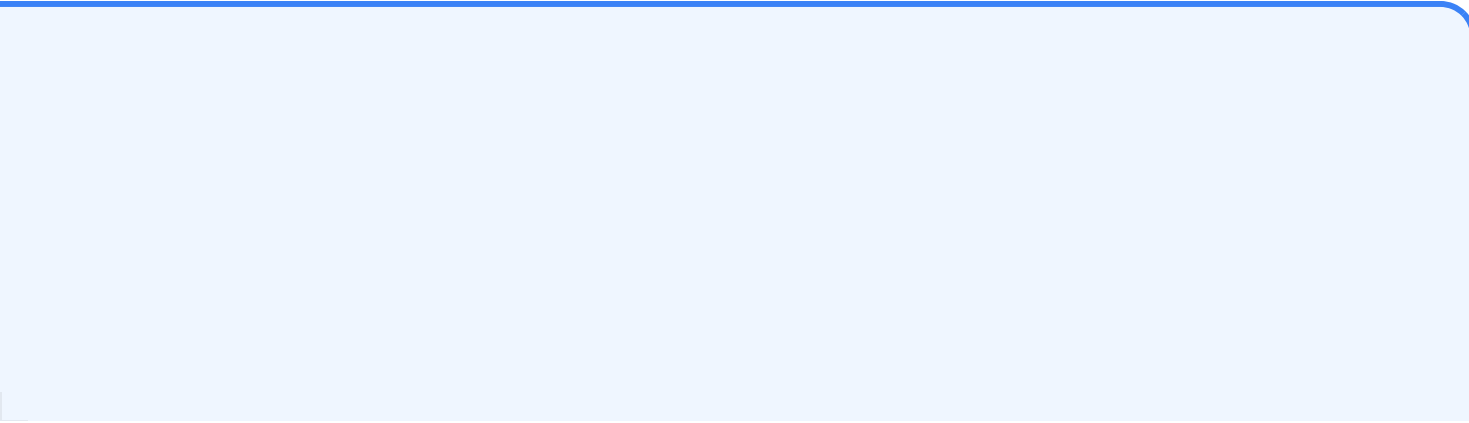
Exponent laws are shortcuts for
multiplication — **add** exponents
subtract when dividing.

THE THREE LAWS YOU USE MOST

$$a^m \times a^n = a^{m+n}$$

5

**ts for counting repeated
nents when multiplying,**



$$a^m \div a^n = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

Negative exponent = reciprocal: $a^{-n} = 1/a^n$

WORKED EXAMPLE

Simplify: $x^5 \times x^3 \div x^2$

$$x^5 \times x^3 = x^{(5+3)} = x^8 \leftarrow \text{add exponents}$$

$$x^8 \div x^2 = x^{(8-2)} = \mathbf{x^6} \leftarrow \text{subtract exponent}$$

⚠ WATCH OUT FOR

ts

MISTAKE

$$(2^3)^4 = 2^7$$

MISTAKE

$$3^{-2} = -9$$



SELF-CHECK

Can you write all seven laws from memory? Can you check your work without looking?

FIX

Multiply exponents for a power of a power: 2^{12}

FIX

Negative exponent = reciprocal: $1/9$

Can you tell apart $2^3 \times 2^4$ from $(2^3)^4$ without